

TIME SERIES ANALYSIS & CLASSIFICATION

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Chapter III: Models for Stationary Time Series

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INTRODUCTION

Models for Stationary Time Series

- ▶ **Recall:** In the last chapter, we used regression to detrend time series data with the hope of removing non-stationary patterns and producing residuals that resembled a stationary process.
- ▶ In this chapter, we consider (linear) time series models for stationary processes. Recall that stationary time series are those whose statistical properties do not change over time.

Terminology and General Linear Process

- **Terminology:** Suppose $\{e_t\}$ is a zero mean white noise process with $\text{Var}(e_t) = \sigma_e^2$. A general linear process is defined by:

$$Y_t = e_t + \Psi_1 e_{t-1} + \Psi_2 e_{t-2} + \Psi_3 e_{t-3} + \dots$$

That is, Y_t , the value of the process at time t , is a weighted linear combination of white noise terms at the current and past times.

- The processes we examine in this chapter are special cases of this general linear process. In general, $\mathbb{E}(Y_t) = 0$ and

$$\gamma_k = \text{Cov}(Y_t, Y_{t-k}) = \sigma_e^2 \sum_{i=0}^{\infty} \Psi_i \Psi_{i+k} \text{ for } k \geq 0,$$

where we set $\Psi_0 = 1$.

Assumptions for Stationarity

- ▶ For mathematical reasons (to ensure stationarity), we will assume that the Ψ_i 's are square summable, that is, $\sum_{i=1}^{\infty} \Psi_i^2 < \infty$.
- ▶ A nonzero mean μ could be added to the right-hand side of the general linear process above; this would not affect the stationarity properties of $\{Y_t\}$. Therefore, there is no harm in assuming that the process $\{Y_t\}$ has a zero mean.

Backshift Operator

Terminology: We define the operator B to satisfy

$$BY_t = Y_{t-1},$$

that is, B "backs up" the current value Y_t one time unit to Y_{t-1} . For this reason, we call B the backshift operator. Similarly,

$$B^2 Y_t = BBY_t = BY_{t-1} = Y_{t-2}$$

In general,

$$B^k Y_t = Y_{t-k}$$

Backshift Operator

- For example, using this new notation, we can rewrite the model

$$Y_t = \frac{1}{3}(e_{t-2} + e_{t-1} + e_t)$$

in the following way:

$$Y_t = \frac{1}{3}(B^2 e_t + B e_t + e_t) \quad \Rightarrow \quad Y_t = \frac{1}{3}(B^2 + B + 1)e_t.$$

MOVING AVERAGE PROCESSES ($MA(q)$)

Terminology: Moving Average Process

Terminology: Suppose $\{e_t\}$ is a zero mean white noise process with $\text{Var}(e_t) = \sigma_e^2$. The process

$$\begin{aligned} Y_t &= e_t - \theta_1 e_{t-1} - \theta_2 e_{t-2} - \cdots - \theta_q e_{t-q} \\ &= e_t - \sum_{i=1}^q \theta_i e_{t-i}, \quad \theta_q \neq 0, \end{aligned}$$

is called a moving average process of order q , denoted by $\text{MA}(q)$.

- It is a special case of the general linear process with $\Psi_0 = 1$, $\Psi_1 = -\theta_1$, $\Psi_2 = -\theta_2$, $\dots$, $\Psi_q = -\theta_q$, and $\Psi_{q^*} = 0$ for all $q^* > q$.

$MA(1)$ Process: Mean and Variance

- ▶ With $q = 1$, the moving average process becomes

$$Y_t = e_t - \theta e_{t-1}$$

known as an $MA(1)$ process. For this process:

- ▶ The mean: $E(Y_t) = E(e_t - \theta e_{t-1}) = E(e_t) - \theta E(e_{t-1}) = 0$
- ▶ The variance:

$$\begin{aligned}\gamma_0 &= \text{Var}(Y_t) = \text{Var}(e_t - \theta e_{t-1}) \\ &= \text{Var}(e_t) + \theta^2 \text{Var}(e_{t-1}) - 2\theta \text{Cov}(e_t, e_{t-1}) \\ &= \sigma_e^2 + \theta^2 \sigma_e^2 = \sigma_e^2(1 + \theta^2)\end{aligned}$$

$MA(1)$ Process: Autocovariance function

- The autocovariance at lag 1:

$$\begin{aligned}\gamma_1 &= \text{Cov}(Y_t, Y_{t-1}) \\ &= \text{Cov}(e_t - \theta e_{t-1}, e_{t-1} - \theta e_{t-2}) \\ &= \text{Cov}(e_t, e_{t-1}) - \theta \text{Cov}(e_t, e_{t-2}) - \theta \text{Cov}(e_{t-1}, e_{t-1}) + \theta^2 \text{Cov}(e_{t-1}, e_{t-2}) \\ &= -\theta \text{Var}(e_{t-1}) \\ &= -\theta \sigma_e^2\end{aligned}$$

$MA(1)$ Process: Autocovariance function

- For any lag $k > 1$, $\gamma_k = \text{Cov}(Y_t, Y_{t-k}) = 0$ because no white noise subscripts in Y_t and Y_{t-k} will overlap.

$$\gamma_k = \begin{cases} \sigma_e^2(1 + \theta^2), & \text{for } k = 0 \\ -\theta\sigma_e^2, & \text{for } k = 1 \\ 0, & \text{for } k > 1 \end{cases}$$

MA(1) Process: Autocorrelation function

Autocorrelation Function for MA(1) process:

$$\rho_k = \frac{\gamma_k}{\gamma_0} = \begin{cases} 1, & \text{for } k = 0 \\ -\theta/(1 + \theta^2), & \text{for } k = 1 \\ 0, & \text{for } k > 1 \end{cases}$$

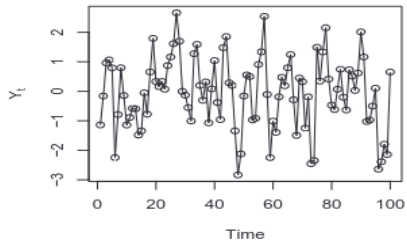
Important Observations

- ▶ The $MA(1)$ process has zero correlation beyond lag $k = 1$.
- ▶ Observations one time unit apart are correlated, but observations more than one time unit apart are not.
- ▶ This fact is crucial in modeling real data using empirical evidence (e.g., sample autocorrelations r_k , etc.).

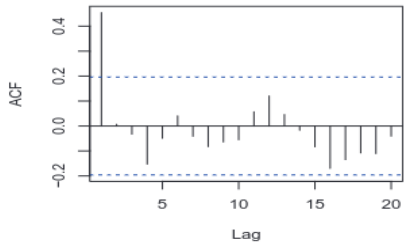
Facts about $MA(1)$ Process

- ▶ When $\theta = 0$, the $MA(1)$ process reduces to a white noise process.
- ▶ As θ ranges from -1 to 1 , the (population) lag 1 autocorrelation ρ_1 ranges from 0.5 to -0.5 .
- ▶ The largest ρ_1 can be is 0.5 (when $\theta = -1$), and the smallest ρ_1 can be is -0.5 (when $\theta = 1$). Therefore, observing a sample lag 1 autocorrelation r_1 outside $[-0.5, 0.5]$ would be inconsistent with the $MA(1)$ model.
- ▶ The population lag 1 autocorrelation $\rho_1 = -\theta/(1 + \theta^2)$ remains the same if θ is replaced by $1/\theta$. Thus, you cannot uniquely identify the corresponding value of θ given the value of ρ_1 for an $MA(1)$ process.

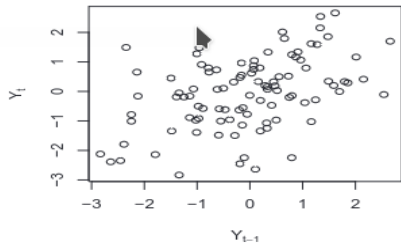
MA(1) simulation



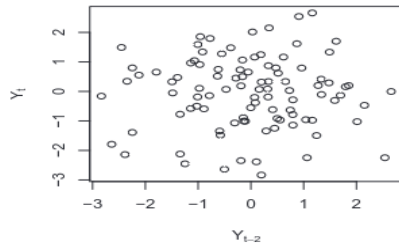
Sample ACF



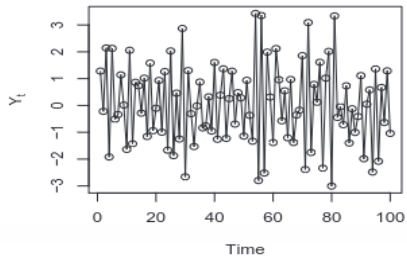
Lag 1 scatterplot



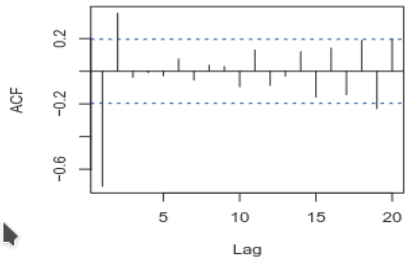
Lag 2 scatterplot



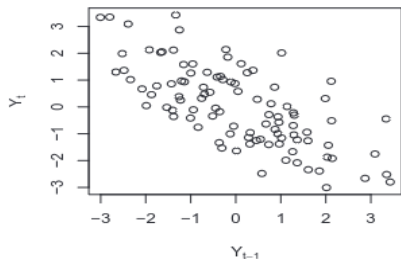
MA(2) simulation



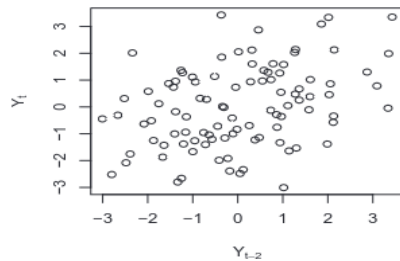
Sample ACF



Lag 1 scatterplot



Lag 2 scatterplot



$MA(q)$ Process

- ▶ The $MA(q)$ process is given by:

$$Y_t = e_t - \theta_1 e_{t-1} - \theta_2 e_{t-2} - \cdots - \theta_q e_{t-q}$$

- ▶ In backshift operator notation, we can write

$$Y_t = (1 - \theta_1 B - \theta_2 B^2 - \cdots - \theta_q B^q) e_t = \Theta(B) e_t,$$

- ▶ Standard calculations show that:

$$E(Y_t) = 0 \quad \text{and} \quad \gamma_0 = \text{Var}(Y_t) = \sigma_e^2(1 + \theta_1^2 + \theta_2^2 + \cdots + \theta_q^2)$$

$MA(q)$ Process: Autocorrelation Function

- ▶ Autocorrelation function for an $MA(q)$ process:

$$\rho_k = \begin{cases} 1, & k = 0 \\ \frac{-\theta_k + \theta_1\theta_{k+1} + \dots + \theta_{q-k}\theta_q}{1 + \theta_1^2 + \theta_2^2 + \dots + \theta_q^2}, & k = 1, 2, \dots, q-1 \\ \frac{-\theta_q}{1 + \theta_1^2 + \theta_2^2 + \dots + \theta_q^2}, & k = q \\ 0, & k > q \end{cases}$$

- ▶ **Salient Feature:** The (population) ACF ρ_k is nonzero for lags $k = 1, 2, \dots, q$. For all lags $k > q$, the ACF $\rho_k = 0$.

AUTOREGRESSIVE PROCESSES ($AR(p)$)

Autoregressive Processes

- Suppose $\{e_t\}$ is a zero-mean white noise process with $\text{Var}(e_t) = \sigma_e^2$. The process

$$\begin{aligned} Y_t &= \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \phi_p Y_{t-p} + e_t \\ &= \sum_{i=1}^p \phi_i Y_{t-i} + e_t, \quad \phi_p \neq 0, \end{aligned}$$

is called an autoregressive process of order p , denoted by $AR(p)$.

- The value of the process at time t , Y_t , is a weighted linear combination of the values of the process from the previous p time points plus a "shock" or "innovation" term e_t at time t .

Autoregressive Processes

- ▶ We assume that e_t , the innovation at time t , is independent of all past process values Y_{t-1} , Y_{t-2} , and so on. That is

$$\text{Cov}(Y_s, e_t) = 0, \quad \forall t, \text{ and } 1 \leq s < t$$

- ▶ We continue to assume that $E(Y_t) = 0$.

$AR(1)$ Process: Terminology

- ▶ Take $p = 1$ in the general $AR(p)$ process and we get $Y_t = \phi Y_{t-1} + e_t$. This is an $AR(1)$ process.
- ▶ Note that if $\phi = 1$, this process reduces to a random walk. If $\phi = 0$, this process reduces to white noise.

AR(1) Process: Variance

- ▶ Assuming stationarity, the variance of Y_t is given by

$$\text{Var}(Y_t) = \text{Var}(\phi Y_{t-1} + e_t) = \phi^2 \text{Var}(Y_{t-1}) + \text{Var}(e_t) = \phi^2 \text{Var}(Y_{t-1}) + \sigma_e^2.$$

- ▶ Assuming stationarity, $\text{Var}(Y_t) = \text{Var}(Y_{t-1}) = \gamma_0$. Therefore,

$$\gamma_0 = \phi^2 \gamma_0 + \sigma_e^2 \Rightarrow \gamma_0 = \frac{\sigma_e^2}{1 - \phi^2}$$

- ▶ Because $\gamma_0 > 0$, this implies $0 < \phi^2 < 1$, that is, $-1 < \phi < 1$.

$AR(1)$ Process: Autocovariance Function

- ▶ To find the autocovariance function γ_k , multiply both sides of the $AR(1)$ equation by Y_{t-k} and take expectation of both sides

$$E[Y_t Y_{t-k}] = E[\phi Y_{t-1} Y_{t-k} + e_t Y_{t-k}]$$

- ▶ Because $\{Y_t\}$ is a zero-mean process (by assumption), we have $\gamma_k = \text{Cov}(Y_t, Y_{t-k}) = E(Y_t Y_{t-k}) - E(Y_t)E(Y_{t-k}) = E(Y_t Y_{t-k})$.

$AR(1)$ Process: Autocovariance Function

- ▶ From these observations, for an $AR(1)$ process, we establish the following (recursive) relationship: $\gamma_k = \phi\gamma_{k-1}$.
 - When $k = 1$, $\gamma_1 = \phi\gamma_0 = \phi(\sigma_e^2/(1 - \phi^2))$.
 - When $k = 2$, $\gamma_2 = \phi\gamma_1 = \phi^2(\sigma_e^2/(1 - \phi^2))$.
- ▶ This pattern continues for larger k . In general, the autocovariance function for an $AR(1)$ process is

$$\gamma_k = \phi^k(\sigma_e^2/(1 - \phi^2)), \quad k = 0, 1, 2, \dots$$

$AR(1)$ Process: Autocovariance Function

- For an $AR(1)$ process,

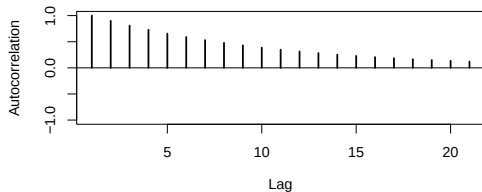
$$\rho_k = \gamma_k / \gamma_0 = \phi^k, \quad k = 0, 1, 2, \dots$$

AR(1) Process: Autocovariance Function

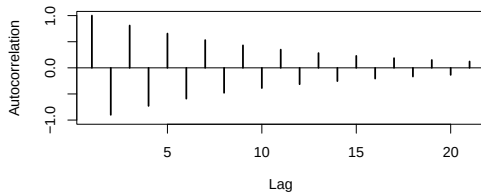
- ▶ **Important Note:** For an AR(1) process, because $-1 < \phi < 1$, the (population) autocorrelation function $\rho_k = \phi^k$ decays exponentially as k increases.
- ▶ Considerations:
 - If ϕ is close to 1, the decay will be slower.
 - If ϕ is not close to 1, the decay will be fast.
 - If $\phi > 0$, all autocorrelations will be positive.
 - If $\phi < 0$, the autocorrelations will alternate between negative and positive.
- ▶ Remember these theoretical patterns to make informed decisions about potential model selection when examining sample ACFs (from real data).

$AR(1)$ Process

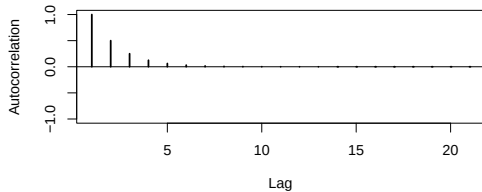
Population ACF, $\phi = 0.9$



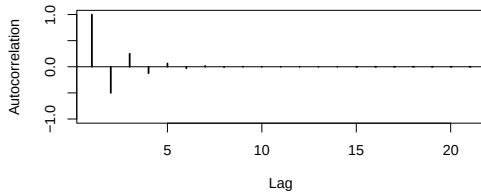
Population ACF, $\phi = -0.9$



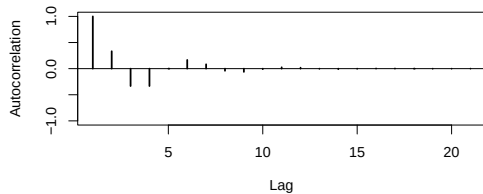
Population ACF, $\phi = 0.5$



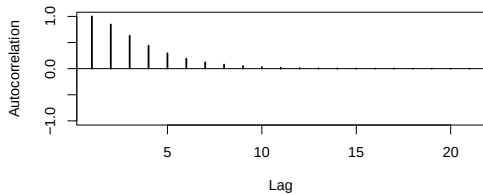
Population ACF, $\phi = -0.5$



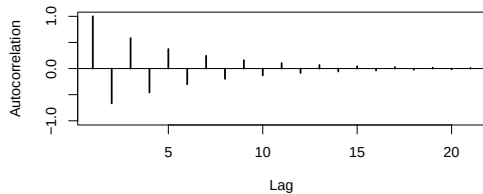
Population ACF for AR(2), with $(\phi_1, \phi_2) = (0.5, -0.5)$



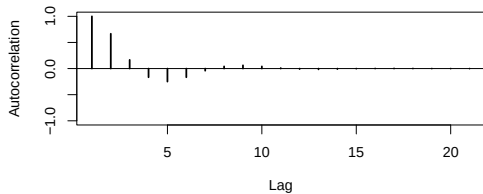
Population ACF for AR(2), with $(\phi_1, \phi_2) = (1.1, -0.3)$



Population ACF for AR(2), with $(\phi_1, \phi_2) = (-0.5, 0.25)$



Population ACF for AR(2), with $(\phi_1, \phi_2) = (1, -0.5)$



$AR(p)$ Process

- Suppose $\{e_t\}$ is a zero mean white noise process with $(e_t) = \sigma_e^2$. The general autoregressive process of order p , denoted $AR(p)$, is

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + e_t.$$

- In backshift operator notation, we can write the model as

$$\Phi(B) Y_t = (1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p) Y_t = e_t,$$

$AR(p)$ Process: Yule-Walker Equations

- ▶ Assuming stationarity and zero means, consider the $AR(p)$ process equation:

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + e_t$$

- ▶ Multiply both sides by Y_{t-k} to get:

$$Y_t Y_{t-k} = \phi_1 Y_{t-1} Y_{t-k} + \phi_2 Y_{t-2} Y_{t-k} + \dots + \phi_p Y_{t-p} Y_{t-k} + e_t Y_{t-k}$$

- ▶ Taking expectations gives:

$$\gamma_k = \phi_1 \gamma_{k-1} + \phi_2 \gamma_{k-2} + \dots + \phi_p \gamma_{k-p}$$

$AR(p)$ Process: Yule-Walker Equations

- ▶ Dividing through by the process variance γ_0 , we get:

$$\rho_k = \phi_1 \rho_{k-1} + \phi_2 \rho_{k-2} + \dots + \phi_p \rho_{k-p}$$

- ▶ Plugging in $k = 1, 2, \dots, p$, and using the fact that $\rho_k = \rho_{-k}$, we get

$$\begin{cases} \rho_1 &= \phi_1 + \phi_2 \rho_1 + \phi_3 \rho_2 + \dots + \phi_p \rho_{p-1} \\ \rho_2 &= \phi_1 \rho_1 + \phi_2 + \phi_3 \rho_1 + \dots + \phi_p \rho_{p-2} \\ &\vdots \\ \rho_p &= \phi_1 \rho_{p-1} + \phi_2 \rho_{p-2} + \phi_3 \rho_{p-3} + \dots + \phi_p \end{cases}$$

$AR(p)$ Process: Yule-Walker Equations

- ▶ Rewriting all the equations together yields

$$\underbrace{\begin{pmatrix} \rho_1 \\ \rho_2 \\ \vdots \\ \rho_{p-1} \\ \rho_p \end{pmatrix}}_{\boldsymbol{\rho}} = \underbrace{\begin{pmatrix} \rho_0 & \rho_1 & \rho_2 & \cdots & \rho_{p-2} & \rho_{p-1} \\ \rho_1 & \rho_0 & \rho_1 & \cdots & \rho_{p-3} & \rho_{p-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \rho_{p-2} & \rho_{p-3} & \rho_{p-4} & \cdots & \rho_0 & \rho_1 \\ \rho_{p-1} & \rho_{p-2} & \rho_{p-3} & \cdots & \rho_1 & \rho_0 \end{pmatrix}}_{\mathbf{R}} \underbrace{\begin{pmatrix} \phi_1 \\ \phi_2 \\ \vdots \\ \phi_{p-1} \\ \phi_p \end{pmatrix}}_{\boldsymbol{\Phi}}.$$

- ▶ These are the Yule-Walker equations. For known values of $\phi_1, \phi_2, \dots, \phi_p$, we can compute the first lag p autocorrelations $\rho_1, \rho_2, \dots, \rho_p$.

$AR(p)$ Process: Yule-Walker Equations

- ▶ Values of ρ_k for $k > p$ can be obtained by using the recursive relation above.
- ▶ The $AR(p)$ ACF tails off as k gets larger. It does so as a mixture of exponential decays and/or damped sine waves, depending on if roots are real or complex.

$AR(p)$ Process: Yule-Walker Equations

- ▶ Recalling that $\rho_0 = 1$, the above equations can be written as

$$\rho = \mathbf{R}\Phi$$

- ▶ Note that this is a well-posed system (with a square coefficients matrix $\mathbf{R}$), i.e., with the same number of constraints (equations, $\mathbf{R}$'s rows) as unknowns (the elements ϕ_j of the unknown vector Φ). Further, $\mathbf{R}$ is full-rank and symmetric, so that invertibility is guaranteed,

$$\Phi = \mathbf{R}^{-1}\rho$$

INVERTIBILITY AND STATIONARITY

Invertibility

- ▶ **Terminology:** We define a process $\{Y_t\}$ to be invertible if it can be written as a "mathematically meaningful" autoregressive process (possibly of infinite order). Invertibility is an important theoretical property.
- ▶ For prediction purposes, it is important to restrict our attention to the class of invertible models.

Invertibility

- ▶ From the definition, we see that stationary autoregressive models are automatically invertible.
- ▶ However, moving average models may not be. For example, consider the $MA(1)$ model

$$Y_t = e_t - \theta e_{t-1}$$

or, slightly rewritten,

$$e_t = Y_t + \theta e_{t-1}$$

Note that we can write

$$e_t = Y_t + \theta(Y_{t-1} + \theta e_{t-2}) = Y_t + \theta Y_{t-1} + \theta^2 e_{t-2}.$$

Repeated similar substitution reveals that

$$e_t = Y_t + \theta Y_{t-1} + \theta^2 Y_{t-2} + \theta^3 Y_{t-3} + \dots,$$

Invertibility

or slightly rewritten,

$$Y_t = -\theta Y_{t-1} - \theta^2 Y_{t-2} - \theta^3 Y_{t-3} - \dots + e_t.$$

- ▶ For this autoregressive representation to be "mathematically meaningful", we need the infinite series of θ coefficients to be finite; that is, we need $\sum_{j=1}^{\infty} \theta^j < \infty$. This occurs if and only if $|\theta| < 1$.
- ▶ We have expressed an $MA(1)$ as an infinite-order AR model.
- ▶ The $MA(1)$ process is invertible if and only if $|\theta| < 1$.

Invertibility

or slightly rewritten,

$$Y_t = -\theta Y_{t-1} - \theta^2 Y_{t-2} - \theta^3 Y_{t-3} - \dots + e_t.$$

- ▶ For this autoregressive representation to be "mathematically meaningful", we need the infinite series of θ coefficients to be finite; that is, we need $\sum_{j=1}^{\infty} \theta^j < \infty$. This occurs if and only if $|\theta| < 1$.
- ▶ We have expressed an $MA(1)$ as an infinite-order AR model.
- ▶ The $MA(1)$ process is invertible if and only if $|\theta| < 1$.

Invertibility

Importance: A model must be invertible for us to be able to identify the model parameters associated with it. For example, for an $MA(1)$ model, it is straightforward to show that both of the following processes have the same autocorrelation function:

$$Y_t = e_t - \theta e_{t-1}$$

$$Y_t = e_t - \frac{1}{\theta} e_{t-1}.$$

Invertibility

- ▶ Put another way, if we knew the common ACF, we could not say if the $MA(1)$ model parameter was θ or $1/\theta$. Thus, we impose the condition that $|\theta| < 1$ to ensure invertibility (identifiability).
- ▶ Note that under this condition, the second $MA(1)$ model, rewritten as

$$Y_t = -\theta Y_{t-1} + \theta^2 Y_{t-2} + \theta^3 Y_{t-3} + \dots + e_t,$$

is no longer meaningful because the series $\sum_{j=1}^{\infty} \theta^j$ diverges.

Invertibility

Generalization: The $MA(q)$ process

$$Y_t = e_t - \theta_1 e_{t-1} - \theta_2 e_{t-2} - \cdots - \theta_q e_{t-q} = (1 - \theta_1 B - \theta_2 B^2 - \cdots - \theta_q B^q) e_t$$

is invertible if and only if the roots of the $MA(q)$ characteristic polynomial

$$\Theta(x) = 1 - \theta_1 x - \theta_2 x^2 - \cdots - \theta_q x^q$$

all exceed 1 in absolute value (or modulus).

Stationarity

- ▶ We have seen that for an $MA(q)$ process has $E(Y_t) = 0$, and the autocorrelation can be written as:

$$\rho_k = \begin{cases} 1, & k = 0 \\ \frac{-\theta_k + \theta_1\theta_{k+1} + \dots + \theta_{q-k}\theta_q}{1 + \theta_1^2 + \theta_2^2 + \dots + \theta_q^2}, & k = 1, 2, \dots, q-1 \\ \frac{-\theta_q}{1 + \theta_1^2 + \theta_2^2 + \dots + \theta_q^2}, & k = q \\ 0, & k > q \end{cases}$$

- ▶ $MA(q)$ models are **always stationary**.

Stationarity

- In order to analyse the stationarity of an $AR(p)$ process we consider the $AR(p)$ characteristic equation

$$\Phi(x) = 1 - \phi_1 x - \phi_2 x^2 - \dots - \phi_p x^p = 0.$$

An $AR(p)$ process is stationary if and only if the p roots of $\Phi(x)$ each exceed 1 in absolute value (or in modulus if the roots are complex).

Stationarity

- Consider an $AR(1)$ process

$$Y_t = \phi Y_{t-1} + e_t \iff (1 - \phi B) Y_t = e_t.$$

The $AR(1)$ characteristic polynomial is $\Phi(x) = 1 - \phi x$. Therefore,

$$\Phi(x) = 1 - \phi x = 0 \implies x = \frac{1}{\phi}.$$

- Clearly,

$$|x| = \frac{1}{\phi} > 1 \iff |\phi| < 1,$$

which was the stated stationarity condition for the $AR(1)$ process.

Stationarity

- Consider an $AR(2)$ process

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + e_t \iff (1 - \phi_1 B - \phi_2 B^2) Y_t = e_t.$$

The $AR(2)$ characteristic polynomial is $\Phi(x) = 1 - \phi_1 x - \phi_2 x^2$ whose two roots are

$$x = \frac{\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2}}{-2\phi_2}$$

- The $AR(2)$ process is stationary if and only if both roots are larger than 1 in absolute value (or in modulus if complex). That is, both roots must lie outside the unit circle.

Summary

We have discussed two important theoretical properties of autoregressive (AR) and moving average (MA) models, namely, stationarity and invertibility. Here is a summary of the important findings.

- ▶ For an $AR(p)$ process to be stationary, we need the roots of the AR characteristic polynomial $\Phi(x) = 1 - \phi_1x - \phi_2x^2 - \dots - \phi_px^p$ to all exceed 1 in absolute value (or modulus).
- ▶ For an $MA(q)$ process to be invertible, we need the roots of the MA characteristic polynomial $\Theta(x) = 1 - \theta_1x - \theta_2x^2 - \dots - \theta_qx^q$ to all exceed 1 in absolute value (or modulus).

Summary

- ▶ All invertible MA processes are stationary.
- ▶ All stationary AR processes are invertible.
- ▶ Any invertible $MA(q)$ process corresponds to an infinite-order AR process.
- ▶ Any stationary $AR(p)$ process corresponds to an infinite-order MA process.

AUTOREGRESSIVE MOVING AVERAGE (ARMA) PROCESSES

Autoregressive Moving Average (ARMA) Processes

Terminology: Suppose $\{e_t\}$ is a zero mean white noise process with $\text{Var}(e_t) = \sigma_e^2$. The process

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + e_t - \theta_1 e_{t-1} - \theta_2 e_{t-2} - \dots - \theta_q e_{t-q}$$

is an autoregressive moving average process of orders p and q , written $ARMA(p, q)$ ($\phi_p \neq 0, \theta_q \neq 0$).

Autoregressive Moving Average (*ARMA*) Processes

Backshift Notation: The $ARMA(p, q)$ process, expressed using backshift notation, is

$$(1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p) Y_t = (1 - \theta_1 B - \theta_2 B^2 - \dots - \theta_q B^q) e_t,$$

or, more succinctly, as

$$\Phi(B) Y_t = \Theta(B) e_t,$$

where

$$\Phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p,$$

$$\Theta(B) = 1 - \theta_1 B - \theta_2 B^2 - \dots - \theta_q B^q.$$

- $\Phi(B)$ and $\Theta(B)$ cannot share any common factors.

Autoregressive Moving Average (*ARMA*) Processes

- ▶ For the $ARMA(p, q)$ process to be stationary, we need the roots of the AR characteristic polynomial $\Phi(x) = 1 - \phi_1x - \phi_2x^2 - \dots - \phi_px^p$ to all exceed 1 in absolute value (or modulus).
- ▶ For the $ARMA(p, q)$ process to be invertible, we need the roots of the MA characteristic polynomial $\Theta(x) = 1 - \theta_1x - \theta_2x^2 - \dots - \theta_qx^q$ to all exceed 1 in absolute value (or modulus).

Remark

$\text{AR}(p)$ and $\text{MA}(q)$ processes are each special cases of the $\text{ARMA}(p, q)$ process.

- ▶ An $\text{ARMA}(p, 0)$ process is the same as an $\text{AR}(p)$ process.
- ▶ An $\text{ARMA}(0, q)$ process is the same as an $\text{MA}(q)$ process.

EXAMPLES: MODEL WRITING AND ANALYSIS

Example

Write each of the models:

1. $Y_t = 0.5 Y_{t-1} + e_t - 0.3 e_{t-1} + 1.2 e_{t-2},$

2. $Y_t = 0.4 Y_{t-1} + 0.45 Y_{t-2} + e_t + e_{t-1} + 0.25 e_{t-2},$

using backshift notation and determine whether the model is stationary and/or invertible.

Example

1. The model $Y_t = 0.5Y_{t-1} + e_t - 0.3e_{t-1} + 1.2e_{t-2}$ is an $ARMA(1, 2)$ with $\phi_1 = 0.5$, $\theta_1 = 0.3$, and $\theta_2 = -1.2$.
 - ▶ In backshift notation, this model is $(1 - 0.5B)Y_t = (1 - 0.3B + 1.2B^2)e_t$.
 - ▶ The AR characteristic polynomial has the root $x = 2$, making the process stationary.
 - ▶ The MA characteristic polynomial has roots $x \approx 0.125 \pm 0.904i$, and the modulus of each root is less than 1. Therefore, this process is not invertible.

Example

2. The model $Y_t = 0.4Y_{t-1} + 0.45Y_{t-2} + e_t + e_{t-1} + 0.25e_{t-2}$, at first glance, appears to be an $ARMA(2, 2)$ with $\phi_1 = 0.4$, $\phi_2 = 0.45$, $\theta_1 = -1$, and $\theta_2 = -0.25$.

- In backshift notation, this model is written as

$$(1 - 0.4B - 0.45B^2)Y_t = (1 + B + 0.25B^2)e_t.$$

- However, the AR and MA characteristic polynomials in this instance factor as

$$(1 + 0.5B)(1 - 0.9B)Y_t = (1 + 0.5B)(1 + 0.5B)e_t.$$

Example

- ▶ In *ARMA* models, the *AR* and *MA* characteristic polynomials cannot share any common factors. Here, they do; namely, $(1 + 0.5B)$.
- ▶ Canceling, we have

$$(1 - 0.9B)Y_t = (1 + 0.5B)e_t,$$

which we identify as an *ARMA*(1,1) model with $\phi_1 = 0.9$ and $\theta_1 = -0.5$.

- ▶ This process is stationary since the root of $\Phi(x) = 1 - 0.9x$ is $x = 1.111 > 1$.
- ▶ This process is invertible since the root of $\Theta(x) = 1 + 0.5x$ is $x = -2$, which exceeds 1 in absolute value.